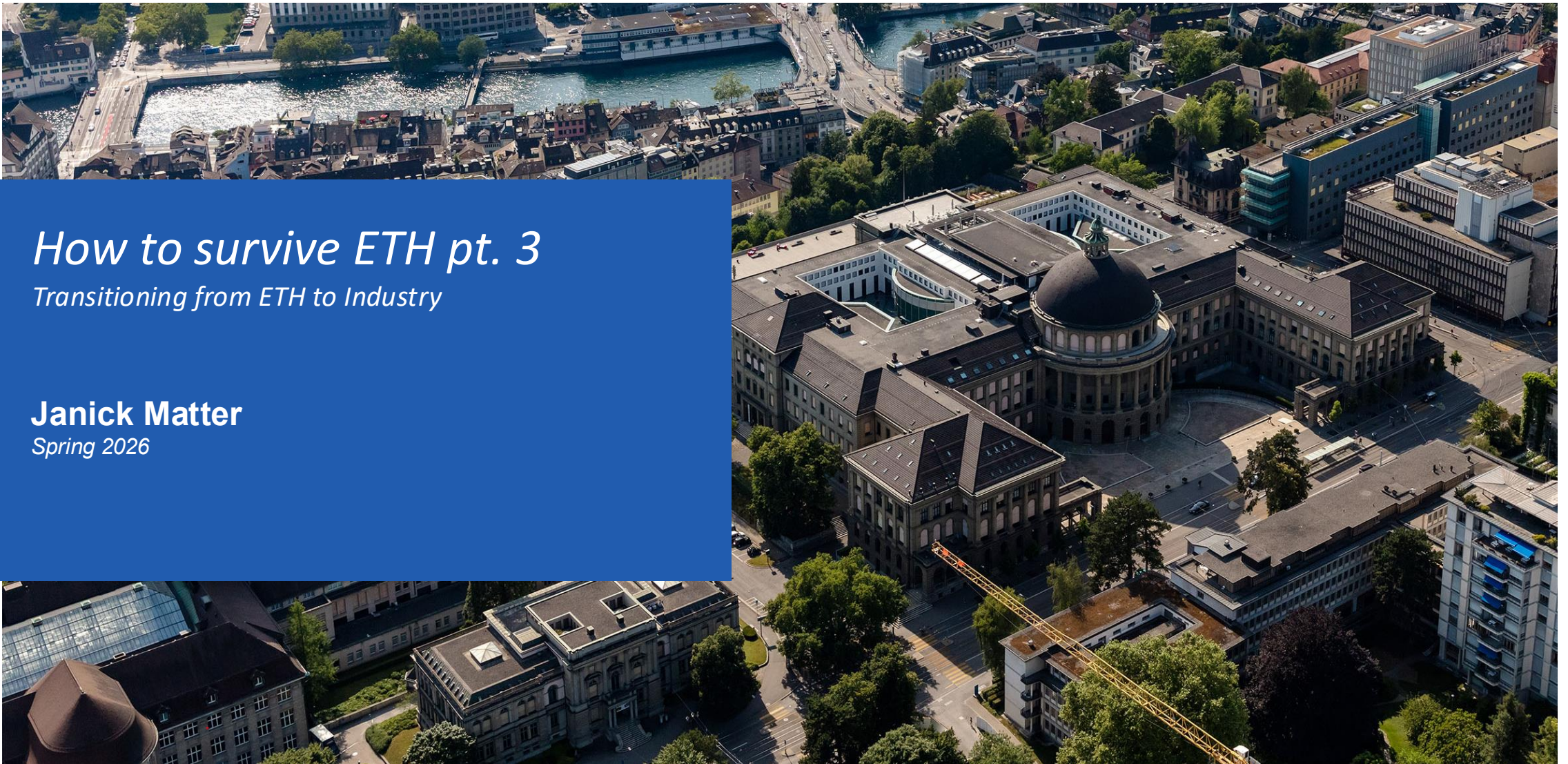


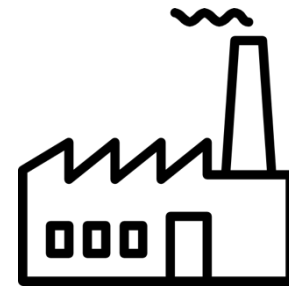
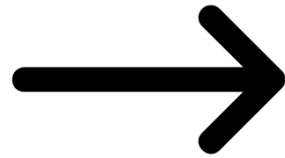
How to survive ETH pt. 3

Transitioning from ETH to Industry

Janick Matter
Spring 2026



ETH zürich

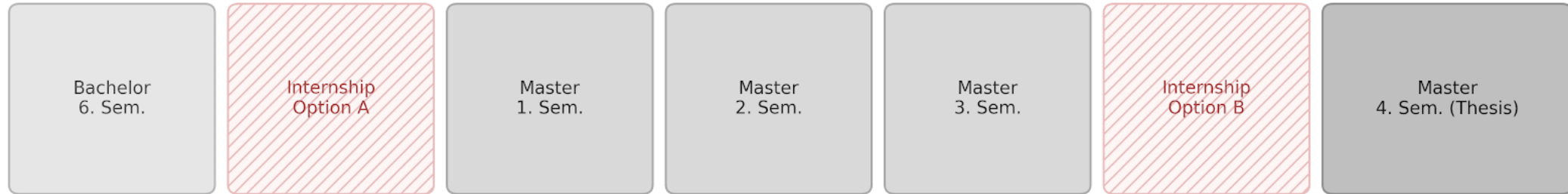


Disclaimer.

This presentation is for informational purposes only and does not constitute legal advice; always review all ETH guidelines and contracts yourself carefully.

*Your **Studies.***

Your studies: timeline & happenings



Internship in industry (12 ECTS)

- Optional (but recommended!)
- Min. 3mts full-time (Basically, 6mts)
- ["Arbeitsbestätigung"](#)
- [Report about the internship](#)

Semester Thesis (12 ECTS)

- Basically a second bachelor thesis (written report & presentation)
- Published in ETH Foyer/hallways, [SIROP](#), or lab websites
- Could also be done in industry

Master Thesis (30 ECTS)

- Only possible once completed semester thesis & passed all core subjects
- 6mts full-time thesis incl. oral presentation and written report
- Must be supervised by a prof listed [HERE](#)
- Can be conducted in industry (!!)

Internship in Industry.

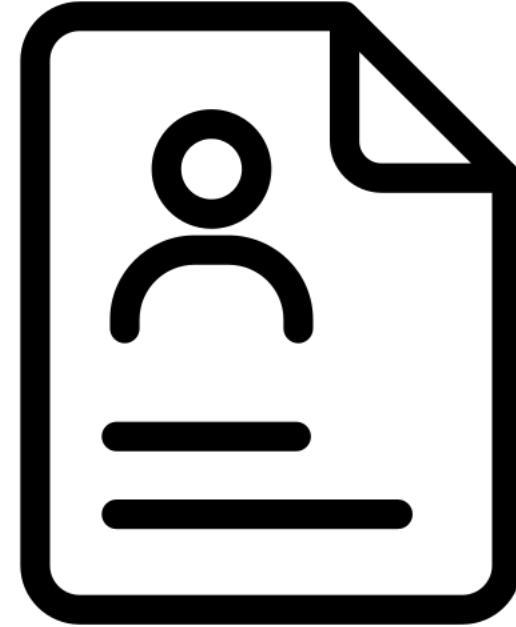
Internship: Application

Apply smart

- Use your existing network (everybody does)
- Network at (ETH) job events and target ppl
 - Polymesse, AMIV Kontakt, VIS Kontaktparty, Campus Interview, AMIV Industry Talks, ETH career center,...
- Cold write (recruiting staff, team leader,...)
- Apply/Reach out through company website
- Apply through job platforms (set alerts on LinkedIn)

What you should prepare (NOW)

- CV (max 2 pages, incl. photo, links & **KEY WORDS**)
- Letter of motivation (template, then adjust)
 - personalized
- Transcripts & References
- A LinkedIn Profile



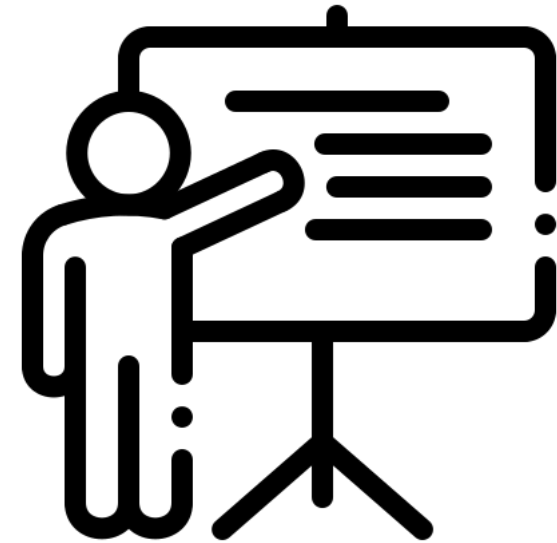
Internship: Interview

How to Interview

- Prepare (for) good questions (both ways)
- Introduction slides*
 - Photos to your CV
 - Recent Projects & Interests (can be technical)
 - “Why you, Why me”
 - Questions
- Think about the soft side of an interview

And what to do once you started

- Take notes of/after meetings
- Keep records of who you’ve met and what you’ve done → Arbeitszeugnis
- Build up network/relationships
- Keep doors open
- Ask (!)



Master Thesis in Industry.

Why in industry?

Why in industry?



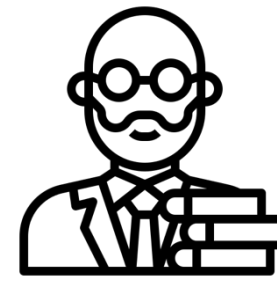
You (Student)

- Real-world problems with direct impact
- Access to proprietary data/systems
- Strong career experience on CV
- Strong networking opportunities during 6mts.
- Possible return offer
- Possible financial compensation (expenses)



PhD / Supervisor

- Potential Industry network
- Potential follow-ups (projects or employment)



Professor

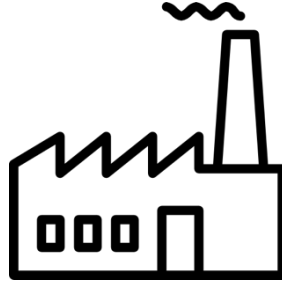
- Strengthens lab-industry network & visibility

Why in industry?



Company supervisor

- Extra capacity on a defined project **or**
- Opportunity to explore new project ideas **with**
- low risk, commitment and cost
- Fresh (and young) perspectives in their team
- Potential recruitment pipeline
- Potential follow-up project/theses
- Visibility of themselves within the company
- ...



Company

- Talent Pipeline (Access to “top ETH talents”)
- Low-cost R&D or concept validation
- Knowledge Transfer: ensuring alignment with cutting-edge technologies
- Brand visibility
- Network expansion (to labs and other parties)
- ...

*Why **not** in industry?*

*Why **not** in industry?*



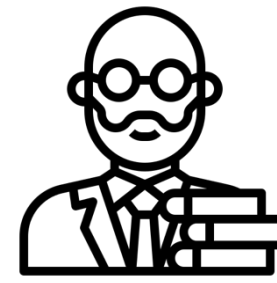
You (Student)

- Limited publication possibilities
- Risk of lower scientific contribution (important for potential PhD)
- Dependence on more parties
- More complex (legal) setup



PhD / Supervisor

- Higher coordination and supervising effort
- Limited publication opportunities
- Less control over project direction and quality
- Likely no contribution to (& citation of) own research
- No financial compensation (per default)



Professor

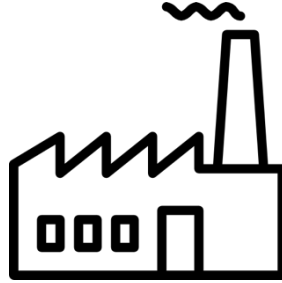
- Higher coordination, administrative and legal effort
- Potential publication conflicts
- Potential IP/legal conflicts
- Limited ability for rigorous supervision
- Risk to academic standards of thesis
- No financial compensation (per default)

Why **not** in industry?



Company supervisor

- Marginal (time & resources) investment
- Coordination effort
- Output uncertainty

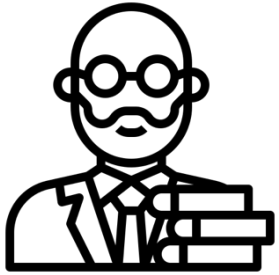


Company

- Coordination effort (& legal risks)
- Output uncertainty

The different options.

Different options



Via (specific) Lab

Some labs (especially in Energy science) have existing collaborations and pre-defined theses topics in collaboration with industry partners.

Application directly through lab/PhDs or through <https://sirop.org/app/>

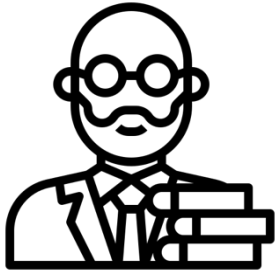


Exploration Lab (EXL)

6-month “innovation traineeship”/6 weeks deep dive (master thesis)
On-site collaboration at partnering (!) company
4:1 mentoring by ETH “innovation engineers”
Interdisciplinary student teams

Application: <https://exl.ethz.ch/apply>

Different options



Via (specific) Lab

“Limited/fixed” topics/fields
“Limited” companies
Given Lab/supervisor

Simple contract setup

ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

HITACHI

Design and Comparative Analysis of Single-Phase and Three-Phase DAB Converters

At Hitachi Energy Research, a design and optimization tool for power electronic converters is being developed, serving different applications and business units. In this thesis, you will be focusing on comparing single-phase and three-phase Dual Active Bridge (DAB) converters considering multiple objectives such as losses, cost, weight, and volume. The analysis will examine the performance of key components, i.e., semiconductors, capacitors, and magnetics under different operating conditions, with a particular emphasis on magnetic design, which is critical for achieving high efficiency and a compact size. This thesis aims to provide insights into selecting the most suitable topology for different scenarios, supporting design decisions at our business units. As part of the thesis, the existing framework will be further developed to support systematic trade-off analysis and enable generic design recommendations for different applications, while also providing the student with the opportunity to gain hands-on experience in converter design and optimization.

Work Description

- 40 % Modelling/implementation
- 20 % Theory
- 40 % Analysis

Prerequisites

- Interest and knowledge in power electronic systems
- Interest and knowledge in magnetics

Supervision: Yunni Li (Hitachi Energy Research)
Location: Baden-Daettwil, Aargau, Switzerland
Contact: Please send your CV and transcript to yunni.li@hitachienergy.com
Professor: Prof. Dr. J. Biela

swissgrid

BUCHER
hydraulics

ABB



HITACHI



Different options



Exploration Lab (EXL)

- Limited spots (around 15)
- Application process (CV, transcripts, letter of motivation, maker portfolio)
- Single start date per year (October)
- Limited Companies (though various projects)
- Limited Supervisors (assigned)
- Non-negotiable Contracts (no compensation for expenses, ...)

Simple contract setup

Different options



Do It Yourself

- Whatever topic/project
- Whenever
- With whatever supervisor you find
- “**Wherever**”

Different options



Do It Yourself

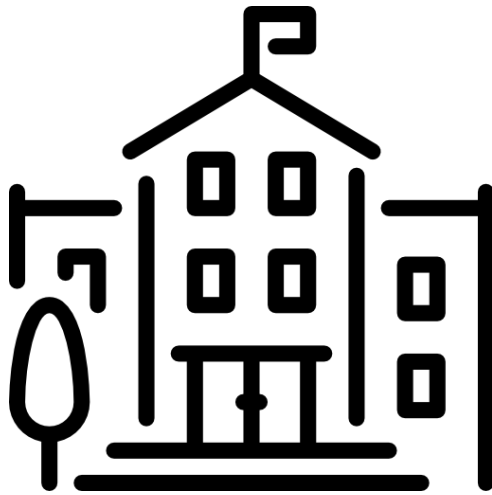


ETH zürich

Eg: <https://jobs.migros.ch/en/career-paths> or <https://www.studyond.com/de/> or <https://careers.axpo.com/jobs/3640276-master-s-bachelor-s-thesis-at-axpo-f-m-d>
Or Zurich

The (official) rules.

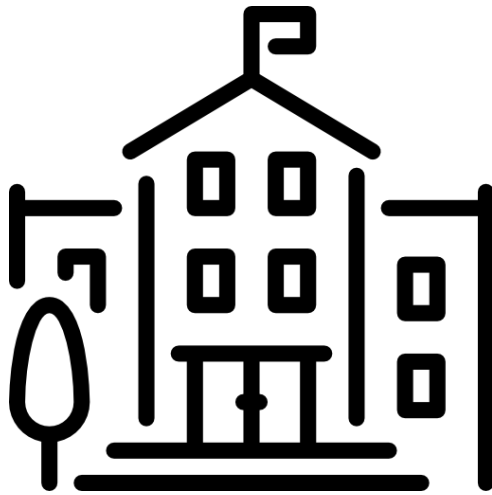
The (official) rules



ETH

- ~~No remuneration~~ (Lohn) for “project work undertaken outside of ETH Zürich [...] which leads to semester papers or Bachelor’s/Master’s theses”
 - ~~Expenses~~ (eg. cost for Food or travel or accommodation) **may be covered** by 3rd parties
- Master theses must be supervised by a professor of the department (incl. adjunct/associated professors)
 - This most often includes a PhD student of that lab

The (official) rules



ETH

- ETH Zurich employees (PhD or Prof) should not sign other NDAs, which have not been checked by the legal counsel. (ETH provides approved NDA template)
- ETH Zurich employees [...] should not sign this, or any other (IP) agreement or contract between the student and the partner.
- Rights to the project results (IP rights) remain with the party who created them (mainly student) if no other agreement is reached between student and partner.

The (official) documents.

The (official) documents



“Declaration Form”

- Who supervises the thesis (ETH vs company)?
- Who provides workspace, data, and resources?
- Where and how is the thesis conducted?
- How is the thesis evaluated and graded?



“3 party NDA ”

- What information is considered confidential?
- Who is allowed to access and share this information?
- How long must confidentiality be maintained?
- What can be published or disclosed?
- How is sensitive information protected and handled?



“2 party (IP) agreement”

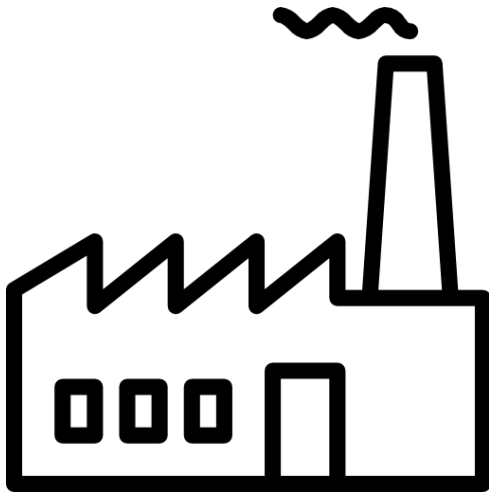
- Who owns the results (IP) of the thesis?
- Can the company use, patent, or commercialize results?
- Can the student publish the work, and under what conditions?
- What support or compensation are provided?

The (official) documents

	Declaration Form (3-party)	NDA (3-party)	Two-Party Agreement
Link	https://ethz.ch/content/dam/ethz/associates/students/studium/zusammenarbeit-externe-partner/Thesis-company-declaration-form-student-partner-eth.docx	https://ethz.ch/content/dam/ethz/associates/students/studium/zusammenarbeit-externe-partner/Thesis-company-3-party-nda-student-partner-eth.docx	https://ethz.ch/content/dam/ethz/associates/students/studium/zusammenarbeit-externe-partner/Thesis-company-2-party-agreement-student-partner.docx
Purpose	<i>Defines framework, roles, supervision, project setup</i>	<i>Protects confidential information</i>	<i>Regulates IP, usage rights, publication</i>
Main focus	<i>Academic requirements, responsibilities</i>	<i>Confidentiality & information handling</i>	<i>Ownership & exploitation of results</i>
IP regulation	<i>Not defined (handled separately)</i>	<i>No IP transfer (only protection)</i>	<i>IP typically assigned to company</i>
Publication	<i>Ensures ETH can access & evaluate thesis</i>	<i>Allows publication of non-confidential content</i>	<i>Publication subject to company review & delay</i>
ETH involvement	<i>Full (ETH is a party)</i>	<i>Full (ETH is a party)</i>	<i>None (ETH not involved)</i>
Signatures	<i>Student + ETH + Company</i>	<i>Student + ETH + Company</i>	<i>Student + Company</i>
Mandatory	<i>Yes (for external thesis setup)</i>	<i>Optional (if confidentiality needed)</i>	<i>Optional (if IP needs regulation, usually standard)</i>
When used	<i>Always for external thesis (no ETH infrastructure necessary)</i>	<i>When company shares sensitive data</i>	<i>When company wants ownership/control of results</i>

*And the (potential) **problems.***

The (potential) problems



We want you, but:

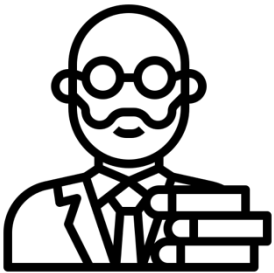
- “We’re not allowed to sign any 3rd party agreements. You (and all supervisors) must sign our agreements.”
- “If you want to do your thesis here you have to sign an employee contract, and by doing so, you have to get paid, is that okay?”
- “We don’t want anyone else to know about your work or results. You can’t publish your thesis.”
- “We need to make sure that all the IP rights of the project stay with us. ”
- ...

The process in reality.

The process



"Why should I supervise a thesis in industry? Not interested."



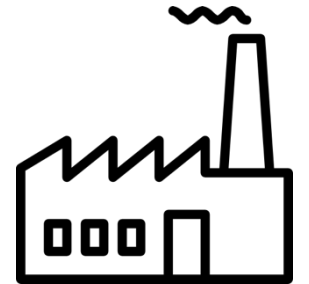
"I don't want any extra effort. Not interested."



"I want to write a thesis in industry! What can I do?"

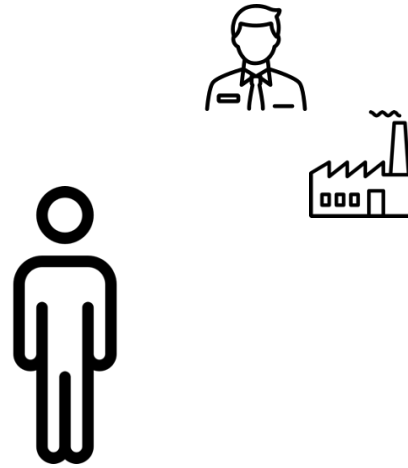


"I want to host/supervise a thesis but no legal risk or large coordination effort."



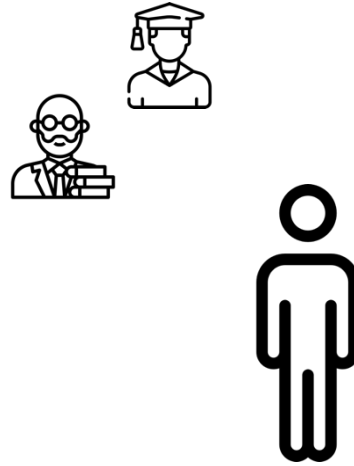
"We want to host students but no legal risk."

The process



1. **Think about topics** which interest you
 - What courses did you like at ETH?
 - Is there a field which you would love to explore more?
2. **Think about companies** which interest you and fit to your fields/s of interest
 - How could your field of interest look like in a project at this company?
 - Who could you reach out to at the company? And how?
3. **Get in touch***

The process



4. **Think about labs/profs** to potentially supervise your thesis (ideally familiar with your topic)

- They must be from this lists:
 - <https://ee.ethz.ch/de/departement/professoren.html>
 - <https://ee.ethz.ch/de/departement/professoren/titularprofessoren.html>
 - https://ethz.ch/content/dam/ethz/special-interest/itet/departement/Studies/D-ITET_Assoziierte_Profs.pdf
- Convince them of your idea, your opportunity and that their effort during 6mts. is limited
 - (show understanding of thesis in industry setup process)

5. **Get in touch with** as many **PhD students** and professors as possible (!)

- Don't do this step sequentially, do it parallel

*The **ideal** scenario.*

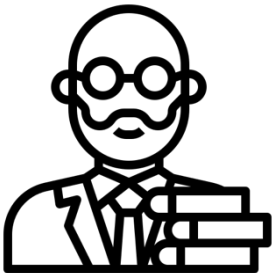
The ideal process



"I have this great idea, let's do it together"



"Great opportunity for me and you & manageable supervision effort. Please sign declaration and NDA"



"Ok, fine."



"Great opportunity for me and you (list pros and give ideas). Please sign declaration and NDA."



"Ok, I don't care about IP rights."



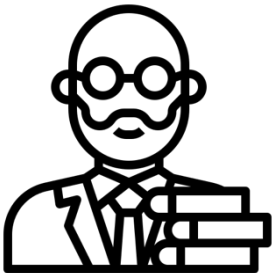
The ideal process



"I have this great idea, let's do it together"



"Great opportunity for me and you & manageable supervision effort. Please sign declaration and NDA"



"Ok, fine."



"Great opportunity for me and you (list pros and give ideas). Please sign declaration and NDA."



"Ok, but what about IP rights?"



"Ok, either we use ETH template, or your contract must comply with ETH regulations"



"Ok, let's use ETH template."



Things can go south.

The process: hickups



"I have this great idea, let's do it together"



"not in my field of research: not interested."

"We cannot sign any company NDA/contracts."

"Ok but the company has to pay us."

"Ok we won't really supervise you. The company must do this."

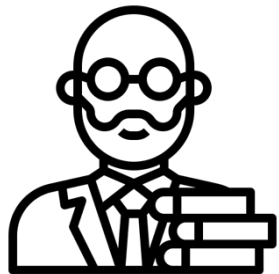
...

"Ok, but we can't sign any 3rd party contracts."

"Ok, but we have nobody to technically supervise you here."

"Ok, but you have to receive a salary."

...



*But you can also **be lucky.***

The process



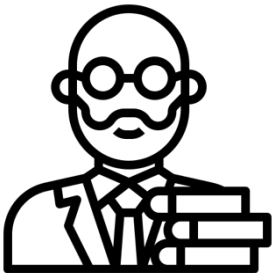
"I have this great idea, let's do it together"



"Oh, you found a company, sure I'll sign their NDA & contracts."

"I don't have much time but we can formally supervise your thesis and just come to your final presentation. just put it on mystudies."

...



"Okay, let me check with our legal department, maybe we can approve ETH contracts (declaration, NDA & 2 party agreement)."

"Yea sure, we can compensate your ETH supervisors, if they sign our versions of contracts"

"Okay, so we will just give you a fixed sum to cover your expenses."

"Okay, we just need an NDA."

...



Probably go for it: (not your fault)

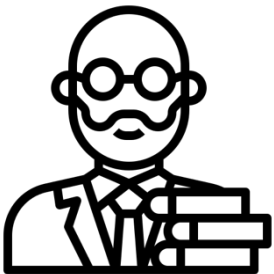
Depends on how comfortable you are & the overall situation: I did not go for this option.

Think about clarifying who the results belong to

The process



"I have this great idea, let's do it together"



"We already have a relationship (and contracts) which govern all topics. You can just sign an approved declaration."



Matchminds: a platform.

Matchminds (under construction)



A platform:

- Companies can publish master theses in different formats on matchminds (similar to sirop)
 - Information provided on the possibilities (for companies and students)
- Students can look and apply for theses
- All contracts are handled through the platform

- Limited number of companies on platform
- Limited projects published
- (no added incentive for eth staff to supervise/support thesis)

Contact: partnership-for-students@ethz.ch

Website: <https://partnerships.ethz.ch>

*Things to **keep** in mind.*

Things to keep in mind



Be aware:

- **Consequences** of incorrect (legal) setup **can be severe (!)**
- Plan enough time **(at least 1-2 months before desired start date)** for the whole setup process to keep things stress free
 - People might response with delays or be on holidays
 - People might need more time/approvals for legal documents
 - ...
- Always look for supervisors (both ETH and company side) in parallel
- Don't wait too long for any reply
 - do "follow-ups" emails
- Plan holidays: 6mts can feel quite long without a break ;)
- If it does not work with someone, there is an alternative!

Do it, try it.

Merci – *For your time and attention*